

Strategy Brief #5

Land Subsidence Challenges in Developing Regions, Gulf of Guinea



Unpacking why land subsidence, driven by groundwater extraction, urbanization, and weak regulation, poses a hidden but escalating threat to infrastructure, livelihoods, and climate resilience across the Gulf of Guinea.

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Introduction

Land subsidence is an under-recognized but critical issue in many developing regions, driven by **groundwater extraction, urbanization, and weak regulation.**



Key Challenge

Subsidence accelerates **flood risk and infrastructure damage**, often without being monitored or managed.

Strategic Insights

- Human activities are a **primary driver of subsidence.**
- Subsidence can exceed **sea-level rise in impact.**
- Monitoring systems are **largely absent in data-scarce regions.**



Strategic Implications

- Hidden risk leading to **unexpected infrastructure failures.**
- Increased vulnerability of already fragile systems.
- Limited awareness among policymakers.



Recommendations

- Establish **subsidence monitoring systems** (e.g., **InSAR**).
- Regulate **groundwater extraction**.
- Integrate subsidence into **risk and planning frameworks**.
- Raise awareness among **decision-makers**.



Value Proposition

Early detection and management reduce **long-term infrastructure and economic losses**.